Akhil **Shelly**

SENIOR ENGINEERING LEAD · SOLUTION ARCHITECT

Toronto, Ontario, Canada

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Professional Summary

Senior Engineering Leader and Solution Architect with 17+ years of experience directing high-impact enterprise initiatives, distributed microservices, and cloud modernization across finance, logistics, and healthcare. Proven track record of bridging executive strategy with production systems, untangling complex legacy monoliths, and fostering high-velocity engineering cultures. Adept at turning around delayed initiatives, driving architectural governance, and delivering fault-tolerant platforms that scale.

Skills

Architecture & Cloud Computing	Solution Architecture, Microsoft/Amazon/Google Cloud, Cloud migrations
Software Development & Engineering	Backend Development – Java, Spring Boot, Python, TypeScript; Frontend & Testing
DevOps & Deployment	CI/CD Pipelines, Version Control & Collaboration Tools
Leadership & Team Management	Leading development teams, onboarding, training, and fostering growth.
Problem-Solving & Research	Technical troubleshooting, solution evaluation
Data Engineering	Data Migration, Data Pipelines, Big Data Technologies
AI	Utilizing AI tools to enhance software development and team productivity

Professional Experience

Avanade

Toronto, ON, Canada

2019 - Present

MANAGER *June 2023 - Present*

- Line manager directing career development, technical evaluations, and promotions for **7 software engineers**.
- Appointed **Bench Ambassador** by practice leadership to steward talent readiness, engineer utilization, and client alignment.
- Led delivery-center focus groups across the **North American Engineering Hubs** analyzing biannual employee survey feedback to drive actionable operational and workplace improvements.
- Standardized technical hiring rubrics across the **Java and Backend Engineering practice**, ensuring consistent, unbiased candidate evaluations.

SENIOR CONSULTANT *Jan 2021 - June 2023*

- Led cross-functional teams through the full SDLC, from requirements gathering to deployment, ensuring seamless integration of microservices.
- Specialized in stabilizing high-risk or delayed projects, resolving complex technical bottlenecks and restoring stakeholder confidence.
- Designed scalable microservice architectures and defined migration roadmaps for moving on-prem legacy systems to the cloud.
- Acted as a technical anchor, onboarding new developers and conducting code reviews to maintain code quality standards.

CONSULTANT *Jan 2019 - Jan 2021*

- Engineered robust backend solutions for industry-leading clients, utilizing Java/Spring Boot and implementing JMS messaging patterns.
- Championed automated testing adoption, writing comprehensive test suites using Cucumber and TypeScript to ensure regression-free releases.
- Maintained CI/CD pipelines ensuring reliable build and deployment processes.

PolicyMedical

Richmond Hill, ON, Canada

SOFTWARE ENGINEER *2016 - 2019*

- Led architectural transition into **Cloud and SaaS engineering**, designing and scaling a multi-tenant healthcare Policy & Contracts platform on **AWS (EC2, S3, RDS)**.
- Designed **REST integration APIs** connecting healthcare regulatory bodies with internal compliance workflows to automate health standards audits.
- Implemented secure B2B authentication and enterprise client network integrations utilizing **SAML SSO** and **AWS VPC/VPN** tunnels.
- Partnered with Client Success to translate customer feedback into high-priority engineering patches and product enhancements.

Evertz Microsystems, Ltd.

Burlington, ON, Canada

DESIGN ENGINEER

2009 - 2016

- Progressed across 7 years to lead engineering on core network management software (**VistaLINK Pro**) monitoring mission-critical broadcast infrastructure worldwide.
- Developed the commercial **Third Party Driver (TPD) Toolkit** standalone product, licensed to tier-1 enterprise broadcast networks (including **Major League Baseball**) for custom SNMP device integration.
- Spearheaded the technical ramp-up of an offshore development team in India, conducting technical training and lead mentorship.

Key Projects

Solution Architect, Senior Software Engineer

Major US Retirement & Financial Services Institution

ADVICE & RETIREMENT PLANNING PLATFORM ***(JAVA, MERCURY COMPOSABLE, REST APIS, REDIS, KAFKA, MICROSERVICES)***

May 2026 - Present

- Stood up net-new cloud-native microservices using the **Mercury Composable** framework following the Strangler Fig pattern to systematically deprecate legacy Spring Boot monoliths.
- Directed service-layer architecture and governance across the Advice workstream, standardizing canonical domain schemas, **REST contracts**, and **Kafka event streams** for digital portals and recordkeeping systems.
- Decoupled complex retirement rules (**IRS contribution limits, lifetime income projections, and decumulation drawdown modeling**) into isolated, composable domain services.
- Partnered directly with Product Managers and **Actuarial SMEs** to translate mathematical modeling specifications into production-ready architectural target states and delivery roadmaps.

Senior Software Engineer, Technical Advisor

Global Logistics Leader

GLOBAL CLEARANCE MODERNIZATION ***(JAVA (J2EE), SPRING BOOT, CLAUDE CODE, GENAI MONOLITH ANALYSIS)***

Jan 2026 - Feb 2026

- Conducted a high-velocity 6-week GenAI advisory engagement utilizing **Claude Code** to reverse-engineer undocumented legacy J2EE customs clearance workflows into structured architectural specs.
- Mapped architectural dependencies and forward-engineering milestones, enabling executive leadership to greenlight the cloud modernization roadmap.

Technical Lead, Solution Architect

Global Premium Financial Services

ENTERPRISE DATA MIGRATION & PRIVACY PLATFORM ***(JAVA, SPRING BOOT, GOOGLE CLOUD PLATFORM, REDIS, APACHE BEAM, PYTHON, CI/CD)***

May 2024 - Nov 2025

- Architected and delivered high-throughput streaming and batch data pipelines processing **tens of millions of records daily with zero data loss**, meeting strict financial audit and PCI compliance standards.
- Led the end-to-end design, implementation, and rollout of **net-new microservices** automating data privacy validation across enterprise data stores.
- Directed a distributed squad of **6–12 cross-functional engineers**, unblocking technical interdependencies and driving continuous delivery.

Technical Lead, Solution Architect

Managed Content Services Provider for Tier-1 US Banks

CLOUD CONTENT MANAGEMENT SYSTEM ***(JAVA, SPRING BOOT, MICROSOFT AZURE, REDIS, AZURE DEVOPS)***

May 2023 - May 2024

- Defined the target-state cloud microservices architecture and API contracts for an enterprise banking content management system hosted on Azure.
- Led technical delivery across two development teams; technical excellence and stakeholder trust **expanded the account into a multi-team delivery engagement driving ~$4M USD in contract growth**.
- Implemented core RESTful microservices and integrated **Redis distributed caching** to optimize document search and metadata retrieval latency.

Senior Software Engineer, Technical Advisor

Big Five Canadian Financial Institution

PERSONAL BANKING SERVICES ***(JAVA, SPRING BOOT, MICROSOFT AZURE, REDIS, ORACLE WEBSPHERE)***

May 2021 - May 2023

- Architected and executed the zero-downtime cloud migration of core retail banking microservices from on-premise WebSphere to Microsoft Azure with **zero customer disruption**.
- Engineered an **Anti-Corruption Layer (ACL)** and **Redis caching layer** to decouple legacy data stores, establishing an architectural pattern subsequently adopted across the retail banking division.
- Served as the divisional cloud migration SME, advising multiple application squads on migration runbooks and automated validation.

Education

McMaster University

Hamilton, ON, Canada

BACHELOR OF ENGINEERING, SPECIALIZED HONOURS IN SOFTWARE ENGINEERING